

Retail Store Inventory Analysis

1. Project Overview

This project analyzes Retail Store Inventory data to improve inventory management, understand sales trends, evaluate product performance, and support business decision-making. The dataset contains information related to products, inventory levels, sales, pricing, demand forecasting, and regional performance. The objective is to identify inventory shortages, optimize stock levels, improve revenue generation, and forecast customer demand using Python, SQL, and Power BI.

2. Dataset Summary

- Rows: 73101 - Columns: 15
- Key Features:
 - ☐ Product details (Product ID, Category)
 - ☐ Inventory information (Inventory Level)
 - ☐ Sales information (Units Sold)
 - ☐ Pricing details (Price, Competitor Pricing)
 - ☐ Regional data (Region)
 - ☐ Seasonal factors (Seasonality)
 - ☐ Demand forecasting
- Promotional information (Holiday / Promotion)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 73100 entries, 0 to 73099
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Date                                  73100 non-null  object
1   Store ID                             73100 non-null  object
2   Product ID                           73100 non-null  object
3   Category                             73100 non-null  object
4   Region                               73100 non-null  object
5   Inventory Level                       73100 non-null  int64
6   Units Sold                           73100 non-null  int64
7   Units Ordered                         73100 non-null  int64
8   Demand Forecast                       73100 non-null  float64
9   Price                                73100 non-null  float64
10  Discount                             73100 non-null  int64
11  Weather Condition                     73100 non-null  object
12  Holiday/Promotion                     73100 non-null  int64
13  Competitor Pricing                    73100 non-null  float64
14  Seasonality                           73100 non-null  object
dtypes: float64(3), int64(5), object(7)
memory usage: 8.4+ MB
```

```

Date          0
Store ID      0
Product ID    0
Category      0
Region        0
Inventory Level 0
Units Sold    0
Units Ordered 0
Demand Forecast 0
Price         0
Discount      0
Weather Condition 0
Holiday/Promotion 0
Competitor Pricing 0
Seasonality   0
dtype: int64

```

- **Missing Data Handling:** Checked for null values and handled missing values in columns such as `inventory_level`, `price`, and `demand_forecast` using median values. Renamed all column names into `snake_case` format for better readability and documentation.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created revenue column by multiplying `units_sold` × `price` to measure total sales performance.
 - Created `inventory_status` column by classifying inventory levels into categories such as Low Stock and Sufficient Stock for inventory monitoring.
 - Created `price_difference` column by calculating the difference between `price` and `competitor_pricing` to evaluate pricing competitiveness.
- **Data Consistency Check:**
 - Verified consistency between **`inventory_level`, `units_sold`, and `demand_forecast`** to identify abnormal values and ensure inventory records aligned with expected sales patterns.
 - Removed duplicate records and validated data types before loading the cleaned dataset into PostgreSQL.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Average Price by Category** – Find the average product price grouped by category.

	Category text	avg double precision
1	Groceries	55.27122168229417
2	Furniture	55.17586094292146
3	Electronics	55.31076303284889
4	Toys	55.03241002526797
5	Clothing	54.88660194174727

2. **Average inventory level by region**– Calculate the average value of price (or inventory-related metric) for each region.

	Region text	avg double precision
1	South	55.04312564901345
2	West	54.9974059036543
3	North	55.31552227342547
4	East	55.18438498010798

3. **Total units sold for each category**– Sum up total units sold for each product category.

	Category text	total_units_sold numeric
1	Groceries	2000482
2	Furniture	2025017
3	Electronics	1960432
4	Toys	1990485
5	Clothing	1999166

4. **Average price by region**– Find the average product price in each region.

	Region text	average_price double precision
1	South	55.04312564901345
2	West	54.9974059036543
3	North	55.31552227342547
4	East	55.18438498010798

5. **Find the average product price in each region.**– Retrieve the five most expensive products in the inventory.

	Date text	Store_ID text	Product_ID text	Category text	Region text	Inventory_Level bigint	Units_Sold bigint	Units_Ordered bigint	Demand_Forecast double precision	Price double precision
1	2022-07-...	S005	P0011	Furniture	North	280	158	173	159.7	
2	2022-11-...	S002	P0001	Clothing	South	464	94	63	89.94	
3	2022-07-...	S002	P0002	Furniture	North	177	79	80	75.71	
4	2022-03-...	S005	P0020	Furniture	South	273	48	152	52.61	
5	2023-03-...	S005	P0002	Furniture	North	406	220	83	212	

6. **Count of products in each category**- Count how many products exist in each category.

	Category text	product_count bigint
1	Groceries	14611
2	Furniture	14699
3	Electronics	14521
4	Toys	14643
5	Clothing	14626

7. **Total revenue by category**– Calculate total revenue (price × units sold) for each category.

	Category text	revenue double precision
1	Groceries	110984538.19999991
2	Furniture	111511804.33999996
3	Electronics	108439061.17999996
4	Toys	109642444.18999967
5	Clothing	109651037.00000083

8. **Products with low inventory**– Find products where inventory level is below 100 units.

	Date text	Store_ID text	Product_ID text	Category text	Region text	Inventory_Level bigint	Units_Sold bigint	Units_Ordered bigint	Demand_Forecast double precision	Price double
1	2022-01-...	S001	P0012	Clothing	West	66	24	70	26.75	
2	2022-01-...	S001	P0013	Toys	South	96	42	85	41.46	
3	2022-01-...	S002	P0011	Groceries	West	67	13	33	7.44	
4	2022-01-...	S002	P0019	Toys	East	67	15	186	34.38	
5	2022-01-...	S003	P0013	Furniture	North	51	17	172	12.86	
6	2022-01-...	S004	P0009	Furniture	North	63	40	73	51.78	
7	2022-01-...	S005	P0010	Clothing	East	87	8	141	14.38	
8	2022-01-...	S005	P0016	Clothing	South	71	11	32	1.85	

9. **Maximum and minimum product prices**– Identify the highest and lowest product prices in the dataset.

	maximum_price double precision	minimum_price double precision
1	100	10

10. **Average demand forecast by season**– Find the average demand forecast grouped by seasonality.

	Seasonality text	average_demand double precision
1	Spring	140.8142496041932
2	Winter	141.90009680065637
3	Summer	140.4700289538377
4	Autumn	142.80340130819602

11. **Total sales by region**- Calculate total units sold in each region.

	Region text	total_sales numeric
1	South	2507799
2	West	2471552
3	North	2484966
4	East	2511265

12. **Category with highest average price**- Identify the category that has the highest average product price.

	Category text	avg_price double precision
1	Electronics	55.31076303284889

13. **Products priced above average price** - List products whose price is greater than the overall average price.

	Date text	Store_ID text	Product_ID text	Category text	Region text	Inventory_Level bigint	Units_Sold bigint	Units_Ordered bigint	Demand_Forecast double precision	Price double precision
1	2022-01-...	S001	P0002	Toys	South	204	150	66	144.04	63.0
2	2022-01-...	S001	P0005	Electronics	East	166	14	135	9.26	73.6
3	2022-01-...	S001	P0006	Groceries	South	138	128	102	139.82	76.8
4	2022-01-...	S001	P0008	Clothing	North	380	312	54	329.73	97.9
5	2022-01-...	S001	P0010	Toys	South	108	28	196	24.47	59.9
6	2022-01-...	S001	P0011	Furniture	South	258	150	153	152.74	58.5

14. **Total records per season-** Count how many records exist for each seasonality type.

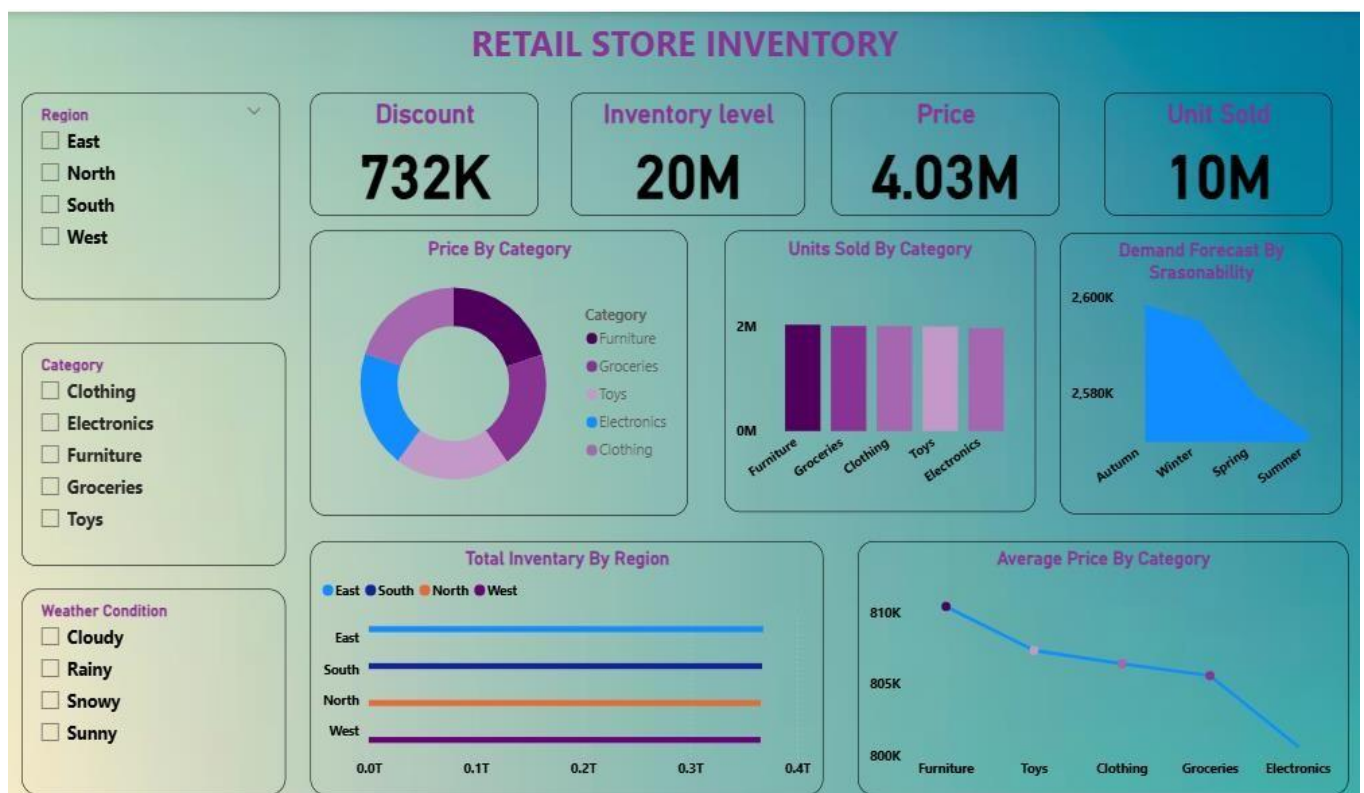
	Seasonality text	total_records bigint
1	Spring	18317
2	Winter	18285
3	Summer	18305
4	Autumn	18193

15. **Find the region having maximum inventory** - Identify the region that holds the highest total inventory level.

	Region text	total_inventory numeric
1	East	5051345

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

1. Improve Stock Allocation by Region

Move more inventory to regions with higher total sales and inventory turnover. Regions with low sales but high stock should have reduced replenishment to avoid overstocking.

2. Avoid Overstocking Low-Demand Products

Products with high inventory but low units sold should be flagged. Reduce purchase orders for these items and consider discounts or bundling to clear stock.

3. Increase Stock for High-Selling Categories

Categories with high total units sold should be prioritized for restocking to avoid stockouts and lost sales opportunities.

4. Optimize Pricing Strategy

If some categories have high average price but low sales, consider promotional pricing or seasonal discounts to improve demand.

5. Use Demand Forecast for Planning

Align procurement with demand forecast trends by season. Increase inventory before peak seasons and reduce during low-demand periods.